

EXPERIMENT-4

Aim

To build an Artificial Neural Network using the Backpropagation algorithm and test its performance using an appropriate dataset.

Procedure

Open Google Colab and create a new notebook.

Import the required Python libraries.

Load the dataset for training and testing.

Preprocess the dataset by normalizing the input values.

Split the dataset into training and testing data.

Create an Artificial Neural Network model.

Train the network using the Backpropagation algorithm.

Test the trained model using test data.

Calculate the accuracy of the model.

Predict the class of new samples.

Program

```
import numpy as np
import pandas as pd

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, classification_report
```

```
iris = load_iris()

X = iris.data
y = iris.target

print("Dataset Loaded Successfully")
```

Dataset Loaded Successfully

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("Dataset Split Successfully")
```

Dataset Split Successfully

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

print("Data Normalization Completed")
```

Data Normalization Completed

```
from sklearn.neural_network import MLPClassifier

model = MLPClassifier(hidden_layer_sizes=(10,10), activation='relu', solver='adam', max_iter=1000, random_state=42)

model.fit(X_train, y_train)

print("ANN Model Trained Successfully")
```

ANN Model Trained Successfully

```
▶ from sklearn.metrics import accuracy_score, classification_report

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy of ANN Model:", accuracy)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Output

```
... Accuracy of ANN Model: 0.9666666666666667
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	0.89	0.94	9
2	0.92	1.00	0.96	11
accuracy			0.97	30
macro avg	0.97	0.96	0.97	30
weighted avg	0.97	0.97	0.97	30

Result: Thus, an Artificial Neural Network was successfully implemented using the Backpropagation algorithm. The model was trained using the Iris dataset and tested successfully with high accuracy for classifying new samples.